



CROSS-LAYER SEEDED FAULT INJECTION ARCHITECTURE

Stress-Testing Every Architectural Layer to Verify Deterministic MCU Safety Containment

🔊 LAYER 1: SENSORY & TRANSDUCTION

- Dropped MIPI camera frames (> 100 ms)
- Optical blinding glare / sudden lux drop
- PTP IEEE-1588 clock skew (> 5 ms)
- Depth map sensor dropout / lens mud

🔧 Camera Drop

💻 LAYER 2: COMPUTE, BUS & MEMORY

- Linux kernel panic / thread crash
- Synthetic DMA memory bus saturation
- `malloc` allocation stalls & page faults
- SPI / CAN-FD CRC checksum corruption

🔧 Kernel Panic

🧠 LAYER 3: MODEL & ALGORITHMIC

- Hallucinated 3D bounding boxes
- Out-of-workspace trajectory proposals
- Discontinuous acceleration jumps
- Adversarial visual perturbation patches

🔧 Hallucination

⚡ LAYER 4: ELECTRICAL & PHYSICAL

- Motor driver over-temperature ($> 105^{\circ}\text{C}$)
- Battery bus voltage sag / brownout (< 10.5 V)
- Sudden external collision load torque
- Encoder phase loss / line break

🔧 Voltage Sag

📋 APPLICATION PROCESSOR (Linux MPU · System 2/1.5)

- Runs Vision Transformers, Deliberation & Action Chunking
- Untrusted, stochastic, prone to crashes and latency tails
- Emits candidate trajectory buffer $\mathbf{p}_t$ over shared IPC link

Candidate $\mathbf{p}_t$

🛡️ REAL-TIME SAFETY ENFORCER (MCU · System 1)

1. **Control Barrier Function Invariant** ($h(x) \geq 0$):
Projects candidate $\mathbf{p}_t$ onto forward invariant safe set $\mathcal{C}$
2. **Dynamic Stopping Clearance Check** ($d_{\text{gap}} > d_{\text{stop}}$):
Vetoes proposals exceeding braking envelope
3. **Zero-Software Hardware Watchdog Monitor:**
Trips Category 1 dynamic stop if MPU heartbeat > 50 ms

Permitted $\mathbf{u}_t$ / E-Stop

⚙️ PHYSICAL PLANT & ACTUATION ($W_t \rightarrow W_{t+1}$)

- Closed-loop motor current loops · Mechanical brakes · Verified safe stop

✅ PASS CRITERION

100% Containment Rate:

- Zero collisions ($d_{\text{gap}} > 0$)
- Bounded stopping time
- Controlled halt < 50 ms

Zero Uncontained Escapes:

Any single uncontained fault trips a **REFUSE** release verdict.

Hardware Watchdog:

Dedicated non-maskable timer operates independently of Linux OS.